

Mini Paper — 2.1 Elements of Computational Thinking

Time: ~30 minutes · Total: 25 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Scenario. *FastPark* is a company building software for a multi-storey car park. Drivers enter at a barrier, take a digital ticket, park, and pay at a machine before leaving. A companion mobile app lets drivers search nearby car parks, see live space availability, and reserve a bay in advance. The software must control the entry/exit barriers, track how many spaces are free, calculate charges, and serve the app.

Q1 [3] (AO2). The developers model the car park abstractly in software.

(a) State what is meant by *abstraction*. [1]

(b) Give **one** real-world detail about the car park that would be *removed* in the model, and **one** detail that must be *kept*, explaining why each choice is appropriate. [2]

Q2 [4] (AO2). The companion app must show "spaces free" for each car park.

(a) Identify **two inputs** the availability feature needs and **two outputs** it produces. [2]

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(b) State **one precondition** that must be true before a driver can reserve a bay, and explain why it is needed. [2]

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Q3 [4] (AO3). Live space-availability data for each car park is requested by many drivers' apps every few seconds.

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Explain how **caching** this data on the server could improve performance, and **evaluate one trade-off** of doing so in this scenario. [4]

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Q4 [4] (AO2). When a driver pays at the exit machine, the software runs through several steps in a fixed order.

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[illegible]

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Discuss which of these tasks could be performed **concurrently**, and evaluate the **benefits and trade-offs** of using concurrent processing here. [6]